Numerical Analysis I

Quiz 9B Name:_____

Question 1.

The number of users of an online service has grown over several years. The table below gives the number of users (in thousands) at selected times.

t	Users (thousands), y
1	3.1
2	5.8
3	9.4
4	12.5
5	16.8
6	20.2
7	24.9

Fit the power-law model

$$y = c_1 t^{c_2}$$

to the data using model linearization.

(a) Linearize the model by taking the natural logarithm of both sides. Express the resulting problem as a linear least squares problem

$$\min_{x \in \mathbb{R}^2} \|Ax - b\|_2.$$

Give the coefficient matrix A and the right hand side b.

(b) We find $x_1 \approx 1.075$ and $x_2 \approx 1.076$. Determine the parameters c_1 and c_2 of the original power-law model.

Question 2.

Find the least squares solution of

$$\min_{x \in \mathbb{R}^2} \|Ax - b\|_2,$$

where

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix}, \qquad b = \begin{bmatrix} 2 \\ 4 \\ 5 \\ 9 \end{bmatrix}.$$

Check that you have obtained the correct answer.